

## 3.9 | Derivatives of Exponential and Logarithmic Functions

### Learning Objectives

- 3.9.1** Find the derivative of exponential functions.
- 3.9.2** Find the derivative of logarithmic functions.
- 3.9.3** Use logarithmic differentiation to determine the derivative of a function.

So far, we have learned how to differentiate a variety of functions, including trigonometric, inverse, and implicit functions. In this section, we explore derivatives of exponential and logarithmic functions. As we discussed in **Introduction to Functions and Graphs**, exponential functions play an important role in modeling population growth and the decay of radioactive materials. Logarithmic functions can help rescale large quantities and are particularly helpful for rewriting complicated expressions.

### Derivative of the Exponential Function

Just as when we found the derivatives of other functions, we can find the derivatives of exponential and logarithmic functions using formulas. As we develop these formulas, we need to make certain basic assumptions. The proofs that these assumptions hold are beyond the scope of this course.

First of all, we begin with the assumption that the function  $B(x) = b^x$ ,  $b > 0$ , is defined for every real number and is continuous. In previous courses, the values of exponential functions for all rational numbers were defined—beginning with the definition of  $b^n$ , where  $n$  is a positive integer—as the product of  $b$  multiplied by itself  $n$  times. Later, we defined  $b^0 = 1$ ,  $b^{-n} = \frac{1}{b^n}$ , for a positive integer  $n$ , and  $b^{s/t} = (\sqrt[t]{b})^s$  for positive integers  $s$  and  $t$ . These definitions leave open the question of the value of  $b^r$  where  $r$  is an arbitrary real number. By assuming the *continuity* of  $B(x) = b^x$ ,  $b > 0$ , we may interpret  $b^r$  as  $\lim_{x \rightarrow r} b^x$  where the values of  $x$  as we take the limit are rational. For example, we may view  $4^\pi$  as the number satisfying

$$4^3 < 4^\pi < 4^4, 4^{3.1} < 4^\pi < 4^{3.2}, 4^{3.14} < 4^\pi < 4^{3.15}, \\ 4^{3.141} < 4^\pi < 4^{3.142}, 4^{3.1415} < 4^\pi < 4^{3.1416}, \dots$$

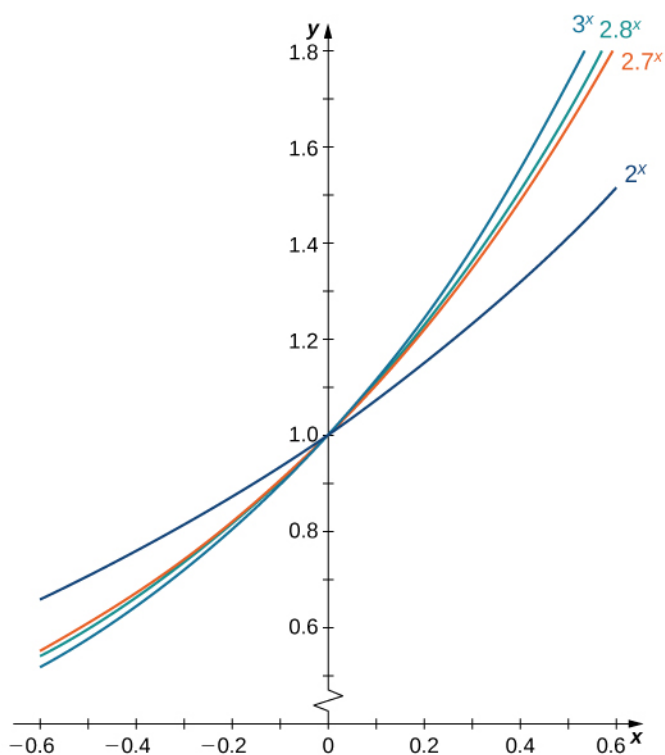
As we see in the following table,  $4^\pi \approx 77.88$ .

| $x$           | $4^x$         | $x$            | $4^x$         |
|---------------|---------------|----------------|---------------|
| $4^3$         | 64            | $4^{3.141593}$ | 77.8802710486 |
| $4^{3.1}$     | 73.5166947198 | $4^{3.1416}$   | 77.8810268071 |
| $4^{3.14}$    | 77.7084726013 | $4^{3.142}$    | 77.9242251944 |
| $4^{3.141}$   | 77.8162741237 | $4^{3.15}$     | 78.7932424541 |
| $4^{3.1415}$  | 77.8702309526 | $4^{3.2}$      | 84.4485062895 |
| $4^{3.14159}$ | 77.8799471543 | $4^4$          | 256           |

**Table 3.6** Approximating a Value of  $4^\pi$

We also assume that for  $B(x) = b^x$ ,  $b > 0$ , the value  $B'(0)$  of the derivative exists. In this section, we show that by making this one additional assumption, it is possible to prove that the function  $B(x)$  is differentiable everywhere.

We make one final assumption: that there is a unique value of  $b > 0$  for which  $B'(0) = 1$ . We define  $e$  to be this unique value, as we did in **Introduction to Functions and Graphs**. **Figure 3.33** provides graphs of the functions  $y = 2^x$ ,  $y = 3^x$ ,  $y = 2.7^x$ , and  $y = 2.8^x$ . A visual estimate of the slopes of the tangent lines to these functions at 0 provides evidence that the value of  $e$  lies somewhere between 2.7 and 2.8. The function  $E(x) = e^x$  is called the **natural exponential function**. Its inverse,  $L(x) = \log_e x = \ln x$  is called the **natural logarithmic function**.



**Figure 3.33** The graph of  $E(x) = e^x$  is between  $y = 2^x$  and  $y = 3^x$ .

For a better estimate of  $e$ , we may construct a table of estimates of  $B'(0)$  for functions of the form  $B(x) = b^x$ . Before doing this, recall that

$$B'(0) = \lim_{x \rightarrow 0} \frac{b^x - b^0}{x - 0} = \lim_{x \rightarrow 0} \frac{b^x - 1}{x} \approx \frac{b^x - 1}{x}$$

for values of  $x$  very close to zero. For our estimates, we choose  $x = 0.00001$  and  $x = -0.00001$  to obtain the estimate

$$\frac{b^{-0.00001} - 1}{-0.00001} < B'(0) < \frac{b^{0.00001} - 1}{0.00001}.$$

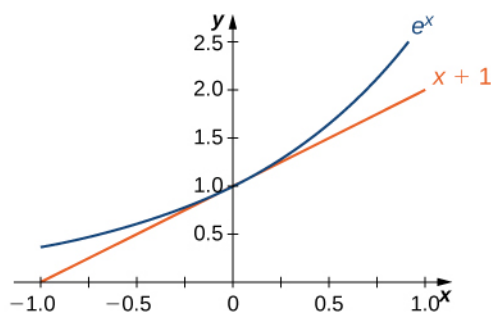
See the following table.

| $b$    | $\frac{b^{-0.00001} - 1}{-0.00001} < B'(0) < \frac{b^{0.00001} - 1}{0.00001}$ | $b$    | $\frac{b^{-0.00001} - 1}{-0.00001} < B'(0) < \frac{b^{0.00001} - 1}{0.00001}$ |
|--------|-------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------|
| 2      | $0.693145 < B'(0) < 0.69315$                                                  | 2.7183 | $1.000002 < B'(0) < 1.000012$                                                 |
| 2.7    | $0.993247 < B'(0) < 0.993257$                                                 | 2.719  | $1.000259 < B'(0) < 1.000269$                                                 |
| 2.71   | $0.996944 < B'(0) < 0.996954$                                                 | 2.72   | $1.000627 < B'(0) < 1.000637$                                                 |
| 2.718  | $0.999891 < B'(0) < 0.999901$                                                 | 2.8    | $1.029614 < B'(0) < 1.029625$                                                 |
| 2.7182 | $0.999965 < B'(0) < 0.999975$                                                 | 3      | $1.098606 < B'(0) < 1.098618$                                                 |

**Table 3.7** Estimating a Value of  $e$

The evidence from the table suggests that  $2.7182 < e < 2.7183$ .

The graph of  $E(x) = e^x$  together with the line  $y = x + 1$  are shown in **Figure 3.34**. This line is tangent to the graph of  $E(x) = e^x$  at  $x = 0$ .



**Figure 3.34** The tangent line to  $E(x) = e^x$  at  $x = 0$  has slope 1.

Now that we have laid out our basic assumptions, we begin our investigation by exploring the derivative of  $B(x) = b^x$ ,  $b > 0$ . Recall that we have assumed that  $B'(0)$  exists. By applying the limit definition to the derivative we conclude that

$$B'(0) = \lim_{h \rightarrow 0} \frac{b^{0+h} - b^0}{h} = \lim_{h \rightarrow 0} \frac{b^h - 1}{h}. \quad (3.28)$$

Turning to  $B'(x)$ , we obtain the following.

$$\begin{aligned}
 B'(x) &= \lim_{h \rightarrow 0} \frac{b^{x+h} - b^x}{h} && \text{Apply the limit definition of the derivative.} \\
 &= \lim_{h \rightarrow 0} \frac{b^x b^h - b^x}{h} && \text{Note that } b^{x+h} = b^x b^h. \\
 &= \lim_{h \rightarrow 0} \frac{b^x(b^h - 1)}{h} && \text{Factor out } b^x. \\
 &= b^x \lim_{h \rightarrow 0} \frac{b^h - 1}{h} && \text{Apply a property of limits.} \\
 &= b^x B'(0) && \text{Use } B'(0) = \lim_{h \rightarrow 0} \frac{b^{0+h} - b^0}{h} = \lim_{h \rightarrow 0} \frac{b^h - 1}{h}.
 \end{aligned}$$

We see that on the basis of the assumption that  $B(x) = b^x$  is differentiable at 0,  $B(x)$  is not only differentiable everywhere, but its derivative is

$$B'(x) = b^x B'(0). \quad (3.29)$$

For  $E(x) = e^x$ ,  $E'(0) = 1$ . Thus, we have  $E'(x) = e^x$ . (The value of  $B'(0)$  for an arbitrary function of the form  $B(x) = b^x$ ,  $b > 0$ , will be derived later.)

### Theorem 3.14: Derivative of the Natural Exponential Function

Let  $E(x) = e^x$  be the natural exponential function. Then

$$E'(x) = e^x.$$

In general,

$$\frac{d}{dx}(e^{g(x)}) = e^{g(x)} g'(x).$$

### Example 3.74

#### Derivative of an Exponential Function

Find the derivative of  $f(x) = e^{\tan(2x)}$ .

#### Solution

Using the derivative formula and the chain rule,

$$\begin{aligned}
 f'(x) &= e^{\tan(2x)} \frac{d}{dx}(\tan(2x)) \\
 &= e^{\tan(2x)} \sec^2(2x) \cdot 2.
 \end{aligned}$$

### Example 3.75

#### Combining Differentiation Rules

Find the derivative of  $y = \frac{e^{x^2}}{x}$ .

**Solution**

Use the derivative of the natural exponential function, the quotient rule, and the chain rule.

$$y' = \frac{(e^{x^2} \cdot 2)x \cdot x - 1 \cdot e^{x^2}}{x^2} \quad \text{Apply the quotient rule.}$$

$$= \frac{e^{x^2}(2x^2 - 1)}{x^2} \quad \text{Simplify.}$$



**3.50** Find the derivative of  $h(x) = xe^{2x}$ .

### Example 3.76

#### Applying the Natural Exponential Function

A colony of mosquitoes has an initial population of 1000. After  $t$  days, the population is given by  $A(t) = 1000e^{0.3t}$ . Show that the ratio of the rate of change of the population,  $A'(t)$ , to the population,  $A(t)$  is constant.

**Solution**

First find  $A'(t)$ . By using the chain rule, we have  $A'(t) = 300e^{0.3t}$ . Thus, the ratio of the rate of change of the population to the population is given by

$$A'(t) = \frac{300e^{0.3t}}{1000e^{0.3t}} = 0.3.$$

The ratio of the rate of change of the population to the population is the constant 0.3.



**3.51** If  $A(t) = 1000e^{0.3t}$  describes the mosquito population after  $t$  days, as in the preceding example, what is the rate of change of  $A(t)$  after 4 days?

## Derivative of the Logarithmic Function

Now that we have the derivative of the natural exponential function, we can use implicit differentiation to find the derivative of its inverse, the natural logarithmic function.

### Theorem 3.15: The Derivative of the Natural Logarithmic Function

If  $x > 0$  and  $y = \ln x$ , then

$$\frac{dy}{dx} = \frac{1}{x}. \quad (3.30)$$

More generally, let  $g(x)$  be a differentiable function. For all values of  $x$  for which  $g'(x) > 0$ , the derivative of

$h(x) = \ln(g(x))$  is given by

$$h'(x) = \frac{1}{g(x)}g'(x). \quad (3.31)$$

### Proof

If  $x > 0$  and  $y = \ln x$ , then  $e^y = x$ . Differentiating both sides of this equation results in the equation

$$e^y \frac{dy}{dx} = 1.$$

Solving for  $\frac{dy}{dx}$  yields

$$\frac{dy}{dx} = \frac{1}{e^y}.$$

Finally, we substitute  $x = e^y$  to obtain

$$\frac{dy}{dx} = \frac{1}{x}.$$

We may also derive this result by applying the inverse function theorem, as follows. Since  $y = g(x) = \ln x$  is the inverse of  $f(x) = e^x$ , by applying the inverse function theorem we have

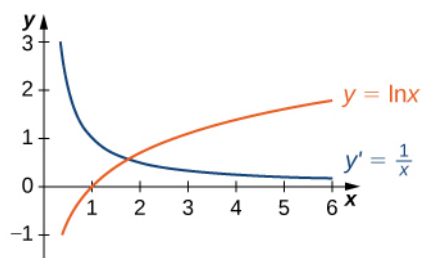
$$\frac{dy}{dx} = \frac{1}{f'(g(x))} = \frac{1}{e^{\ln x}} = \frac{1}{x}.$$

Using this result and applying the chain rule to  $h(x) = \ln(g(x))$  yields

$$h'(x) = \frac{1}{g(x)}g'(x).$$

□

The graph of  $y = \ln x$  and its derivative  $\frac{dy}{dx} = \frac{1}{x}$  are shown in **Figure 3.35**.



**Figure 3.35** The function  $y = \ln x$  is increasing on  $(0, +\infty)$ . Its derivative  $y' = \frac{1}{x}$  is greater than zero on  $(0, +\infty)$ .

## Example 3.77

### Taking a Derivative of a Natural Logarithm

Find the derivative of  $f(x) = \ln(x^3 + 3x - 4)$ .

### Solution

Use **Equation 3.31** directly.

$$\begin{aligned} f'(x) &= \frac{1}{x^3 + 3x - 4} \cdot (3x^2 + 3) && \text{Use } g(x) = x^3 + 3x - 4 \text{ in } h'(x) = \frac{1}{g(x)}g'(x). \\ &= \frac{3x^2 + 3}{x^3 + 3x - 4} && \text{Rewrite.} \end{aligned}$$

## Example 3.78

### Using Properties of Logarithms in a Derivative

Find the derivative of  $f(x) = \ln\left(\frac{x^2 \sin x}{2x + 1}\right)$ .

### Solution

At first glance, taking this derivative appears rather complicated. However, by using the properties of logarithms prior to finding the derivative, we can make the problem much simpler.

$$\begin{aligned} f(x) &= \ln\left(\frac{x^2 \sin x}{2x + 1}\right) = 2\ln x + \ln(\sin x) - \ln(2x + 1) && \text{Apply properties of logarithms.} \\ f'(x) &= \frac{2}{x} + \cot x - \frac{2}{2x + 1} && \text{Apply sum rule and } h'(x) = \frac{1}{g(x)}g'(x). \end{aligned}$$



**3.52** Differentiate:  $f(x) = \ln(3x + 2)^5$ .

Now that we can differentiate the natural logarithmic function, we can use this result to find the derivatives of  $y = \log_b x$  and  $y = b^x$  for  $b > 0$ ,  $b \neq 1$ .

### Theorem 3.16: Derivatives of General Exponential and Logarithmic Functions

Let  $b > 0$ ,  $b \neq 1$ , and let  $g(x)$  be a differentiable function.

i. If  $y = \log_b x$ , then

$$\frac{dy}{dx} = \frac{1}{x \ln b}. \quad (3.32)$$

More generally, if  $h(x) = \log_b(g(x))$ , then for all values of  $x$  for which  $g(x) > 0$ ,

$$h'(x) = \frac{g'(x)}{g(x) \ln b}. \quad (3.33)$$

ii. If  $y = b^x$ , then



$$\frac{dy}{dx} = b^x \ln b. \quad (3.34)$$

More generally, if  $h(x) = b^{g(x)}$ , then

$$h'(x) = b^{g(x)} g'(x) \ln b. \quad (3.35)$$

### Proof

If  $y = \log_b x$ , then  $b^y = x$ . It follows that  $\ln(b^y) = \ln x$ . Thus  $y \ln b = \ln x$ . Solving for  $y$ , we have  $y = \frac{\ln x}{\ln b}$ .

Differentiating and keeping in mind that  $\ln b$  is a constant, we see that

$$\frac{dy}{dx} = \frac{1}{x \ln b}.$$

The derivative in **Equation 3.33** now follows from the chain rule.

If  $y = b^x$ , then  $\ln y = x \ln b$ . Using implicit differentiation, again keeping in mind that  $\ln b$  is constant, it follows that

$\frac{1}{y} \frac{dy}{dx} = \ln b$ . Solving for  $\frac{dy}{dx}$  and substituting  $y = b^x$ , we see that

$$\frac{dy}{dx} = y \ln b = b^x \ln b.$$

The more general derivative (**Equation 3.35**) follows from the chain rule.

□

## Example 3.79

### Applying Derivative Formulas

Find the derivative of  $h(x) = \frac{3^x}{3^x + 2}$ .

### Solution

Use the quotient rule and **Derivatives of General Exponential and Logarithmic Functions**.

$$h'(x) = \frac{3^x \ln 3(3^x + 2) - 3^x \ln 3(3^x)}{(3^x + 2)^2} \quad \text{Apply the quotient rule.}$$

$$= \frac{2 \cdot 3^x \ln 3}{(3^x + 2)^2} \quad \text{Simplify.}$$

## Example 3.80

### Finding the Slope of a Tangent Line

Find the slope of the line tangent to the graph of  $y = \log_2(3x + 1)$  at  $x = 1$ .

### Solution

To find the slope, we must evaluate  $\frac{dy}{dx}$  at  $x = 1$ . Using **Equation 3.33**, we see that

$$\frac{dy}{dx} = \frac{3}{(3x+1)\ln 2}.$$

By evaluating the derivative at  $x = 1$ , we see that the tangent line has slope

$$\left. \frac{dy}{dx} \right|_{x=1} = \frac{3}{4\ln 2} = \frac{3}{\ln 16}.$$



**3.53** Find the slope for the line tangent to  $y = 3^x$  at  $x = 2$ .

## Logarithmic Differentiation

At this point, we can take derivatives of functions of the form  $y = (g(x))^n$  for certain values of  $n$ , as well as functions of the form  $y = b^{g(x)}$ , where  $b > 0$  and  $b \neq 1$ . Unfortunately, we still do not know the derivatives of functions such as  $y = x^x$  or  $y = x^\pi$ . These functions require a technique called **logarithmic differentiation**, which allows us to differentiate any function of the form  $h(x) = g(x)^{f(x)}$ . It can also be used to convert a very complex differentiation problem into a simpler one, such as finding the derivative of  $y = \frac{x\sqrt{2x+1}}{e^x \sin^3 x}$ . We outline this technique in the following problem-solving strategy.

### Problem-Solving Strategy: Using Logarithmic Differentiation

1. To differentiate  $y = h(x)$  using logarithmic differentiation, take the natural logarithm of both sides of the equation to obtain  $\ln y = \ln(h(x))$ .
2. Use properties of logarithms to expand  $\ln(h(x))$  as much as possible.
3. Differentiate both sides of the equation. On the left we will have  $\frac{1}{y} \frac{dy}{dx}$ .
4. Multiply both sides of the equation by  $y$  to solve for  $\frac{dy}{dx}$ .
5. Replace  $y$  by  $h(x)$ .

### Example 3.81

#### Using Logarithmic Differentiation

Find the derivative of  $y = (2x^4 + 1)^{\tan x}$ .

**Solution**

Use logarithmic differentiation to find this derivative.

$$\ln y = \ln(2x^4 + 1)^{\tan x}$$

$$\ln y = \tan x \ln(2x^4 + 1)$$

$$\frac{1}{y} \frac{dy}{dx} = \sec^2 x \ln(2x^4 + 1) + \frac{8x^3}{2x^4 + 1} \cdot \tan x$$

$$\frac{dy}{dx} = y \cdot \left( \sec^2 x \ln(2x^4 + 1) + \frac{8x^3}{2x^4 + 1} \cdot \tan x \right)$$

$$\frac{dy}{dx} = (2x^4 + 1)^{\tan x} \left( \sec^2 x \ln(2x^4 + 1) + \frac{8x^3}{2x^4 + 1} \cdot \tan x \right)$$

Step 1. Take the natural logarithm of both sides.

Step 2. Expand using properties of logarithms.

Step 3. Differentiate both sides. Use the product rule on the right.

Step 4. Multiply by  $y$  on both sides.

Step 5. Substitute  $y = (2x^4 + 1)^{\tan x}$ .

**Example 3.82****Using Logarithmic Differentiation**

Find the derivative of  $y = \frac{x\sqrt{2x+1}}{e^x \sin^3 x}$ .

**Solution**

This problem really makes use of the properties of logarithms and the differentiation rules given in this chapter.

$$\ln y = \ln \frac{x\sqrt{2x+1}}{e^x \sin^3 x}$$

$$\ln y = \ln x + \frac{1}{2} \ln(2x+1) - x \ln e - 3 \ln \sin x$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{x} + \frac{1}{2x+1} - 1 - 3 \frac{\cos x}{\sin x}$$

$$\frac{dy}{dx} = y \left( \frac{1}{x} + \frac{1}{2x+1} - 1 - 3 \cot x \right)$$

$$\frac{dy}{dx} = \frac{x\sqrt{2x+1}}{e^x \sin^3 x} \left( \frac{1}{x} + \frac{1}{2x+1} - 1 - 3 \cot x \right)$$

Step 1. Take the natural logarithm of both sides.

Step 2. Expand using properties of logarithms.

Step 3. Differentiate both sides.

Step 4. Multiply by  $y$  on both sides.

Step 5. Substitute  $y = \frac{x\sqrt{2x+1}}{e^x \sin^3 x}$ .

**Example 3.83****Extending the Power Rule**

Find the derivative of  $y = x^r$  where  $r$  is an arbitrary real number.

**Solution**

The process is the same as in **Example 3.82**, though with fewer complications.

$$\ln y = \ln x^r \quad \text{Step 1. Take the natural logarithm of both sides.}$$

$$\ln y = r \ln x \quad \text{Step 2. Expand using properties of logarithms.}$$

$$\frac{1}{y} \frac{dy}{dx} = r \frac{1}{x} \quad \text{Step 3. Differentiate both sides.}$$

$$\frac{dy}{dx} = y \frac{r}{x} \quad \text{Step 4. Multiply by } y \text{ on both sides.}$$

$$\frac{dy}{dx} = x^r \frac{r}{x} \quad \text{Step 5. Substitute } y = x^r.$$

$$\frac{dy}{dx} = rx^{r-1} \quad \text{Simplify.}$$



**3.54** Use logarithmic differentiation to find the derivative of  $y = x^x$ .



**3.55** Find the derivative of  $y = (\tan x)^\pi$ .

## 3.9 EXERCISES

For the following exercises, find  $f'(x)$  for each function.

331.  $f(x) = x^2 e^x$

332.  $f(x) = \frac{e^{-x}}{x}$

333.  $f(x) = e^{x^3 \ln x}$

334.  $f(x) = \sqrt{e^{2x} + 2x}$

335.  $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

336.  $f(x) = \frac{10^x}{\ln 10}$

337.  $f(x) = 2^{4x} + 4x^2$

338.  $f(x) = 3^{\sin 3x}$

339.  $f(x) = x^\pi \cdot \pi^x$

340.  $f(x) = \ln(4x^3 + x)$

341.  $f(x) = \ln \sqrt{5x - 7}$

342.  $f(x) = x^2 \ln 9x$

343.  $f(x) = \log(\sec x)$

344.  $f(x) = \log_7(6x^4 + 3)^5$

345.  $f(x) = 2^x \cdot \log_3 7^{x^2 - 4}$

For the following exercises, use logarithmic differentiation to find  $\frac{dy}{dx}$ .

346.  $y = x^{\sqrt{x}}$

347.  $y = (\sin 2x)^{4x}$

348.  $y = (\ln x)^{\ln x}$

349.  $y = x^{\log_2 x}$

350.  $y = (x^2 - 1)^{\ln x}$

351.  $y = x^{\cot x}$

352.  $y = \frac{x + 11}{\sqrt[3]{x^2 - 4}}$

353.  $y = x^{-1/2} (x^2 + 3)^{2/3} (3x - 4)^4$

354. **[T]** Find an equation of the tangent line to the graph of  $f(x) = 4xe^{(x^2 - 1)}$  at the point where  $x = -1$ . Graph both the function and the tangent line.

355. **[T]** Find the equation of the line that is normal to the graph of  $f(x) = x \cdot 5^x$  at the point where  $x = 1$ . Graph both the function and the normal line.

356. **[T]** Find the equation of the tangent line to the graph of  $x^3 - x \ln y + y^3 = 2x + 5$  at the point where  $x = 2$ .

(Hint: Use implicit differentiation to find  $\frac{dy}{dx}$ .) Graph both the curve and the tangent line.

357. Consider the function  $y = x^{1/x}$  for  $x > 0$ .

- Determine the points on the graph where the tangent line is horizontal.
- Determine the points on the graph where  $y' > 0$  and those where  $y' < 0$ .

358. The formula  $I(t) = \frac{\sin t}{e^t}$  is the formula for a decaying alternating current.

- a. Complete the following table with the appropriate values.

| $t$              | $\frac{\sin t}{e^t}$ |
|------------------|----------------------|
| 0                | (i)                  |
| $\frac{\pi}{2}$  | (ii)                 |
| $\pi$            | (iii)                |
| $\frac{3\pi}{2}$ | (iv)                 |
| $2\pi$           | (v)                  |
| $\frac{5\pi}{2}$ | (vi)                 |
| $3\pi$           | (vii)                |
| $\frac{7\pi}{2}$ | (viii)               |
| $4\pi$           | (ix)                 |

- b. Using only the values in the table, determine where the tangent line to the graph of  $I(t)$  is horizontal.

359. **[T]** The population of Toledo, Ohio, in 2000 was approximately 500,000. Assume the population is increasing at a rate of 5% per year.

- Write the exponential function that relates the total population as a function of  $t$ .
- Use a. to determine the rate at which the population is increasing in  $t$  years.
- Use b. to determine the rate at which the population is increasing in 10 years.

360. **[T]** An isotope of the element erbium has a half-life of approximately 12 hours. Initially there are 9 grams of the isotope present.

- Write the exponential function that relates the amount of substance remaining as a function of  $t$ , measured in hours.
- Use a. to determine the rate at which the substance is decaying in  $t$  hours.
- Use b. to determine the rate of decay at  $t = 4$  hours.

361. **[T]** The number of cases of influenza in New York City from the beginning of 1960 to the beginning of 1961 is modeled by the function  $N(t) = 5.3e^{0.093t^2 - 0.87t}$ , ( $0 \leq t \leq 4$ ), where  $N(t)$  gives the number of cases (in thousands) and  $t$  is measured in years, with  $t = 0$  corresponding to the beginning of 1960.

- Show work that evaluates  $N(0)$  and  $N(4)$ . Briefly describe what these values indicate about the disease in New York City.
- Show work that evaluates  $N'(0)$  and  $N'(3)$ . Briefly describe what these values indicate about the disease in New York City.

362. **[T]** The *relative rate of change* of a differentiable function  $y = f(x)$  is given by  $\frac{100 \cdot f'(x)}{f(x)}\%$ . One model for population growth is a Gompertz growth function, given by  $P(x) = ae^{-b \cdot e^{-cx}}$  where  $a$ ,  $b$ , and  $c$  are constants.

- Find the relative rate of change formula for the generic Gompertz function.
- Use a. to find the relative rate of change of a population in  $x = 20$  months when  $a = 204$ ,  $b = 0.0198$ , and  $c = 0.15$ .
- Briefly interpret what the result of b. means.

For the following exercises, use the population of New York City from 1790 to 1860, given in the following table.

| Years since 1790 | Population |
|------------------|------------|
| 0                | 33,131     |
| 10               | 60,515     |
| 20               | 96,373     |
| 30               | 123,706    |
| 40               | 202,300    |
| 50               | 312,710    |
| 60               | 515,547    |
| 70               | 813,669    |

**Table 3.8** New York City Population Over Time **Source:** [http://en.wikipedia.org/wiki/Largest\\_cities\\_in\\_the\\_United\\_States\\_by\\_population\\_by\\_decade](http://en.wikipedia.org/wiki/Largest_cities_in_the_United_States_by_population_by_decade).

363. **[T]** Using a computer program or a calculator, fit a growth curve to the data of the form  $p = ab^t$ .

364. **[T]** Using the exponential best fit for the data, write a table containing the derivatives evaluated at each year.

365. **[T]** Using the exponential best fit for the data, write a table containing the second derivatives evaluated at each year.

366. **[T]** Using the tables of first and second derivatives and the best fit, answer the following questions:

- Will the model be accurate in predicting the future population of New York City? Why or why not?
- Estimate the population in 2010. Was the prediction correct from a.?

## CHAPTER 3 REVIEW

### KEY TERMS

**acceleration** is the rate of change of the velocity, that is, the derivative of velocity

**amount of change** the amount of a function  $f(x)$  over an interval  $[x, x + h]$  is  $f(x + h) - f(x)$

**average rate of change** is a function  $f(x)$  over an interval  $[x, x + h]$  is  $\frac{f(x + h) - f(x)}{h}$

**chain rule** the chain rule defines the derivative of a composite function as the derivative of the outer function evaluated at the inner function times the derivative of the inner function

**constant multiple rule** the derivative of a constant  $c$  multiplied by a function  $f$  is the same as the constant multiplied by the derivative:  $\frac{d}{dx}(cf(x)) = cf'(x)$

**constant rule** the derivative of a constant function is zero:  $\frac{d}{dx}(c) = 0$ , where  $c$  is a constant

**derivative** the slope of the tangent line to a function at a point, calculated by taking the limit of the difference quotient, is the derivative

**derivative function** gives the derivative of a function at each point in the domain of the original function for which the derivative is defined

**difference quotient** of a function  $f(x)$  at  $a$  is given by

$$\frac{f(a + h) - f(a)}{h} \text{ or } \frac{f(x) - f(a)}{x - a}$$

**difference rule** the derivative of the difference of a function  $f$  and a function  $g$  is the same as the difference of the derivative of  $f$  and the derivative of  $g$ :  $\frac{d}{dx}(f(x) - g(x)) = f'(x) - g'(x)$

**differentiable at a** a function for which  $f'(a)$  exists is differentiable at  $a$

**differentiable function** a function for which  $f'(x)$  exists is a differentiable function

**differentiable on S** a function for which  $f'(x)$  exists for each  $x$  in the open set  $S$  is differentiable on  $S$

**differentiation** the process of taking a derivative

**higher-order derivative** a derivative of a derivative, from the second derivative to the  $n$ th derivative, is called a higher-order derivative

**implicit differentiation** is a technique for computing  $\frac{dy}{dx}$  for a function defined by an equation, accomplished by differentiating both sides of the equation (remembering to treat the variable  $y$  as a function) and solving for  $\frac{dy}{dx}$

**instantaneous rate of change** the rate of change of a function at any point along the function  $a$ , also called  $f'(a)$ , or the derivative of the function at  $a$

**logarithmic differentiation** is a technique that allows us to differentiate a function by first taking the natural logarithm of both sides of an equation, applying properties of logarithms to simplify the equation, and differentiating implicitly

**marginal cost** is the derivative of the cost function, or the approximate cost of producing one more item

**marginal profit** is the derivative of the profit function, or the approximate profit obtained by producing and selling one more item

**marginal revenue** is the derivative of the revenue function, or the approximate revenue obtained by selling one more item



**population growth rate** is the derivative of the population with respect to time

**power rule** the derivative of a power function is a function in which the power on  $x$  becomes the coefficient of the term and the power on  $x$  in the derivative decreases by 1: If  $n$  is an integer, then  $\frac{d}{dx}x^n = nx^{n-1}$

**product rule** the derivative of a product of two functions is the derivative of the first function times the second function plus the derivative of the second function times the first function:  $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + g'(x)f(x)$

**quotient rule** the derivative of the quotient of two functions is the derivative of the first function times the second function minus the derivative of the second function times the first function, all divided by the square of the second function:  $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - g'(x)f(x)}{(g(x))^2}$

**speed** is the absolute value of velocity, that is,  $|v(t)|$  is the speed of an object at time  $t$  whose velocity is given by  $v(t)$

**sum rule** the derivative of the sum of a function  $f$  and a function  $g$  is the same as the sum of the derivative of  $f$  and the derivative of  $g$ :  $\frac{d}{dx}(f(x) + g(x)) = f'(x) + g'(x)$

## KEY EQUATIONS

- **Difference quotient**

$$Q = \frac{f(x) - f(a)}{x - a}$$

- **Difference quotient with increment  $h$**

$$Q = \frac{f(a+h) - f(a)}{a+h-a} = \frac{f(a+h) - f(a)}{h}$$

- **Slope of tangent line**

$$m_{\tan} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$m_{\tan} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

- **Derivative of  $f(x)$  at  $a$**

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

- **Average velocity**

$$v_{\text{ave}} = \frac{s(t) - s(a)}{t - a}$$

- **Instantaneous velocity**

$$v(a) = s'(a) = \lim_{t \rightarrow a} \frac{s(t) - s(a)}{t - a}$$

- **The derivative function**

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- **Derivative of sine function**

$$\frac{d}{dx}(\sin x) = \cos x$$

- **Derivative of cosine function**

$$\frac{d}{dx}(\cos x) = -\sin x$$

- **Derivative of tangent function**

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

- **Derivative of cotangent function**

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

- **Derivative of secant function**

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

- **Derivative of cosecant function**

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

- **The chain rule**

$$h'(x) = f'(g(x))g'(x)$$

- **The power rule for functions**

$$h'(x) = n(g(x))^{n-1} g'(x)$$

- **Inverse function theorem**

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} \text{ whenever } f'(f^{-1}(x)) \neq 0 \text{ and } f(x) \text{ is differentiable.}$$

- **Power rule with rational exponents**

$$\frac{d}{dx}(x^{m/n}) = \frac{m}{n}x^{(m/n)-1}.$$

- **Derivative of inverse sine function**

$$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$$

- **Derivative of inverse cosine function**

$$\frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}}$$

- **Derivative of inverse tangent function**

$$\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$$

- **Derivative of inverse cotangent function**

$$\frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}$$

- **Derivative of inverse secant function**

$$\frac{d}{dx} \sec^{-1} x = \frac{1}{|x|\sqrt{x^2-1}}$$

- **Derivative of inverse cosecant function**

$$\frac{d}{dx} \csc^{-1} x = \frac{-1}{|x|\sqrt{x^2-1}}$$

- **Derivative of the natural exponential function**

$$\frac{d}{dx}(e^{g(x)}) = e^{g(x)} g'(x)$$

- **Derivative of the natural logarithmic function**

$$\frac{d}{dx}(\ln g(x)) = \frac{1}{g(x)} g'(x)$$

- **Derivative of the general exponential function**

$$\frac{d}{dx}(b^{g(x)}) = b^{g(x)} g'(x) \ln b$$

- **Derivative of the general logarithmic function**

$$\frac{d}{dx}(\log_b g(x)) = \frac{g'(x)}{g(x) \ln b}$$

## KEY CONCEPTS

### 3.1 Defining the Derivative

- The slope of the tangent line to a curve measures the instantaneous rate of change of a curve. We can calculate it by finding the limit of the difference quotient or the difference quotient with increment  $h$ .
- The derivative of a function  $f(x)$  at a value  $a$  is found using either of the definitions for the slope of the tangent line.
- Velocity is the rate of change of position. As such, the velocity  $v(t)$  at time  $t$  is the derivative of the position  $s(t)$  at time  $t$ . Average velocity is given by

$$v_{\text{ave}} = \frac{s(t) - s(a)}{t - a}.$$

Instantaneous velocity is given by

$$v(a) = s'(a) = \lim_{t \rightarrow a} \frac{s(t) - s(a)}{t - a}.$$

- We may estimate a derivative by using a table of values.

### 3.2 The Derivative as a Function

- The derivative of a function  $f(x)$  is the function whose value at  $x$  is  $f'(x)$ .
- The graph of a derivative of a function  $f(x)$  is related to the graph of  $f(x)$ . Where  $f(x)$  has a tangent line with positive slope,  $f'(x) > 0$ . Where  $f(x)$  has a tangent line with negative slope,  $f'(x) < 0$ . Where  $f(x)$  has a horizontal tangent line,  $f'(x) = 0$ .
- If a function is differentiable at a point, then it is continuous at that point. A function is not differentiable at a point if it is not continuous at the point, if it has a vertical tangent line at the point, or if the graph has a sharp corner or cusp.
- Higher-order derivatives are derivatives of derivatives, from the second derivative to the  $n$ th derivative.

### 3.3 Differentiation Rules

- The derivative of a constant function is zero.
- The derivative of a power function is a function in which the power on  $x$  becomes the coefficient of the term and the power on  $x$  in the derivative decreases by 1.
- The derivative of a constant  $c$  multiplied by a function  $f$  is the same as the constant multiplied by the derivative.
- The derivative of the sum of a function  $f$  and a function  $g$  is the same as the sum of the derivative of  $f$  and the derivative of  $g$ .
- The derivative of the difference of a function  $f$  and a function  $g$  is the same as the difference of the derivative of  $f$  and the derivative of  $g$ .
- The derivative of a product of two functions is the derivative of the first function times the second function plus the derivative of the second function times the first function.
- The derivative of the quotient of two functions is the derivative of the first function times the second function minus

the derivative of the second function times the first function, all divided by the square of the second function.

- We used the limit definition of the derivative to develop formulas that allow us to find derivatives without resorting to the definition of the derivative. These formulas can be used singly or in combination with each other.

### 3.4 Derivatives as Rates of Change

- Using  $f(a + h) \approx f(a) + f'(a)h$ , it is possible to estimate  $f(a + h)$  given  $f'(a)$  and  $f(a)$ .
- The rate of change of position is velocity, and the rate of change of velocity is acceleration. Speed is the absolute value, or magnitude, of velocity.
- The population growth rate and the present population can be used to predict the size of a future population.
- Marginal cost, marginal revenue, and marginal profit functions can be used to predict, respectively, the cost of producing one more item, the revenue obtained by selling one more item, and the profit obtained by producing and selling one more item.

### 3.5 Derivatives of Trigonometric Functions

- We can find the derivatives of  $\sin x$  and  $\cos x$  by using the definition of derivative and the limit formulas found earlier. The results are

$$\frac{d}{dx} \sin x = \cos x \quad \frac{d}{dx} \cos x = -\sin x.$$

- With these two formulas, we can determine the derivatives of all six basic trigonometric functions.

### 3.6 The Chain Rule

- The chain rule allows us to differentiate compositions of two or more functions. It states that for  $h(x) = f(g(x))$ ,

$$h'(x) = f'(g(x))g'(x).$$

In Leibniz's notation this rule takes the form

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

- We can use the chain rule with other rules that we have learned, and we can derive formulas for some of them.
- The chain rule combines with the power rule to form a new rule:

$$\text{If } h(x) = (g(x))^n, \text{ then } h'(x) = n(g(x))^{n-1} g'(x).$$

- When applied to the composition of three functions, the chain rule can be expressed as follows: If  $h(x) = f(g(k(x)))$ , then  $h'(x) = f'(g(k(x)))g'(k(x))k'(x)$ .

### 3.7 Derivatives of Inverse Functions

- The inverse function theorem allows us to compute derivatives of inverse functions without using the limit definition of the derivative.
- We can use the inverse function theorem to develop differentiation formulas for the inverse trigonometric functions.

### 3.8 Implicit Differentiation

- We use implicit differentiation to find derivatives of implicitly defined functions (functions defined by equations).
- By using implicit differentiation, we can find the equation of a tangent line to the graph of a curve.

### 3.9 Derivatives of Exponential and Logarithmic Functions

- On the basis of the assumption that the exponential function  $y = b^x$ ,  $b > 0$  is continuous everywhere and

differentiable at 0, this function is differentiable everywhere and there is a formula for its derivative.

- We can use a formula to find the derivative of  $y = \ln x$ , and the relationship  $\log_b x = \frac{\ln x}{\ln b}$  allows us to extend our differentiation formulas to include logarithms with arbitrary bases.
- Logarithmic differentiation allows us to differentiate functions of the form  $y = g(x)^{f(x)}$  or very complex functions by taking the natural logarithm of both sides and exploiting the properties of logarithms before differentiating.

## CHAPTER 3 REVIEW EXERCISES

*True or False?* Justify the answer with a proof or a counterexample.

367. Every function has a derivative.
368. A continuous function has a continuous derivative.
369. A continuous function has a derivative.
370. If a function is differentiable, it is continuous.

Use the limit definition of the derivative to exactly evaluate the derivative.

371.  $f(x) = \sqrt{x+4}$

372.  $f(x) = \frac{3}{x}$

Find the derivatives of the following functions.

373.  $f(x) = 3x^3 - \frac{4}{x^2}$

374.  $f(x) = (4 - x^2)^3$

375.  $f(x) = e^{\sin x}$

376.  $f(x) = \ln(x+2)$

377.  $f(x) = x^2 \cos x + x \tan(x)$

378.  $f(x) = \sqrt{3x^2 + 2}$

379.  $f(x) = \frac{x}{4} \sin^{-1}(x)$

380.  $x^2 y = (y+2) + xy \sin(x)$

Find the following derivatives of various orders.

381. First derivative of  $y = x \ln(x) \cos x$

382. Third derivative of  $y = (3x+2)^2$

383. Second derivative of  $y = 4^x + x^2 \sin(x)$

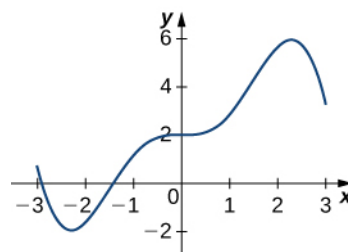
Find the equation of the tangent line to the following equations at the specified point.

384.  $y = \cos^{-1}(x) + x$  at  $x = 0$

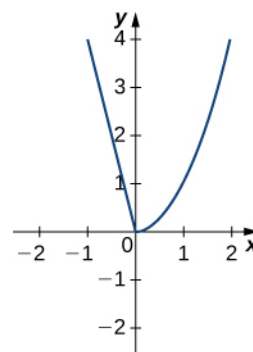
385.  $y = x + e^x - \frac{1}{x}$  at  $x = 1$

Draw the derivative for the following graphs.

386.



387.



The following questions concern the water level in Ocean City, New Jersey, in January, which can be approximated by  $w(t) = 1.9 + 2.9 \cos\left(\frac{\pi}{6}t\right)$ , where  $t$  is measured in hours after midnight, and the height is measured in feet.

388. Find and graph the derivative. What is the physical meaning?

**389.** Find  $w'(3)$ . What is the physical meaning of this value?

The following questions consider the wind speeds of Hurricane Katrina, which affected New Orleans, Louisiana, in August 2005. The data are displayed in a table.

| Hours after Midnight,<br>August 26 | Wind Speed<br>(mph) |
|------------------------------------|---------------------|
| 1                                  | 45                  |
| 5                                  | 75                  |
| 11                                 | 100                 |
| 29                                 | 115                 |
| 49                                 | 145                 |
| 58                                 | 175                 |
| 73                                 | 155                 |
| 81                                 | 125                 |
| 85                                 | 95                  |
| 107                                | 35                  |

**Table 3.9** Wind Speeds of Hurricane Katrina **Source:**  
[http://news.nationalgeographic.com/news/2005/09/0914\\_050914\\_katrina\\_timeline.html](http://news.nationalgeographic.com/news/2005/09/0914_050914_katrina_timeline.html).

**390.** Using the table, estimate the derivative of the wind speed at hour 39. What is the physical meaning?

**391.** Estimate the derivative of the wind speed at hour 83. What is the physical meaning?